Francisco Perez Carbajal

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SUMMARY

Senior Data Scientist with over 7 years of experience applying machine learning algorithms to solve complex business problems, coupled with more than 10 years of academic expertise. Proven track record leading end-to-end data science initiatives across retail and consulting sectors, with deep expertise in forecasting, pricing optimization, and strategic modeling. Specialized in time series forecasting and ensemble methods for demand prediction. Comfortable translating analytical insights into actionable business decisions and working collaboratively in cross-functional teams. My commitment to driving transformative change has led me to contribute significantly to various projects, including advanced forecasting systems, process automation with computer vision, and visual search.

TECHNICAL TOOLKIT:

Machine Learning & AI:

- Algorithms: Regression, classification, clustering, neural networks, deep learning, decision trees, ensemble methods, Random Forest, XGBoost.
- Frameworks**:** Scikit-Learn, Keras, TensorFlow, PyTorch, OpenCV, Prophet.
- NLP **:** Text classification.
- Computer Vision**:** Object detection, semantic segmentation, facial recognition, image similarity.

Programming & Data:

- Languages: Python, R, SQL, Fortran, C#.
- Data Analysis & Visualization: Numpy, Pandas, Matplotlib, Seaborn
- Big Data: PySpark.
- Parallel Processing: Multiprocessing.

Statistical modeling and analysis:

- Time Series Analysis and Forecasting: ARIMA, SARIMA, Prophet, ensamble forecasting methods
- Advanced statistical algorithms, hypothesis testing
- Regression (parametric and non-parametric)
- Bayesian and non-Bayesian methods

Infrastructure & Tools:

- Cloud Platforms: Google Cloud Platform (GCP), IBM Cloud Paks
- Databases: BigQuery, SQL Server, Azure Data Studio, DBeaver
- Platforms: Docker, Jupyter Notebook, Google Colab
- Version Control: Git, GitHub
- Methodologies: SCRUM, Agile
- Software: Matlab, Microsoft Office, SPSS, Linux

PROFESSIONAL EXPERIENCE

SENIOR DATA SCIENTIST, CONSULTANT, AXITY (CLIENT: WALMART MÉXICO); CDMX – AUG 2025 - PRESENT

Automated Checkout Forecasting System for Walmart México

Leading data science efforts in a collaborative team of 5 (2 additional data scientists, 1 ML engineer, 1 project lead) to develop an automated forecasting system for Walmart México's checkout operations. The project leverages GitHub for version control and focuses on predicting purchase volumes using sales and inventory variables to optimize staffing and resource allocation.

Key Contributions:

- Developed and implemented ensemble forecasting models combining ARIMA, SARIMA, Random Forest, XGBoost, and Prophet algorithms to predict demand for perishable goods
- **Achieved measurable improvement** in forecasting accuracy for 8 out of 25 perishable product time series by implementing ensemble methods that outperformed individual models (ARIMA, SARIMA, Random Forest standalone)

- Engineered features from sales and inventory data to enhance prediction accuracy
- Designed and deployed models on Google Cloud Platform (GCP) infrastructure, utilizing BigQuery for data access and cloud storage solutions
- Collaborated in cross-functional team environment, coordinating with data scientists from multiple consulting firms (Axity and K2) and ML engineers

Tools Utilized: Python, Scikit-Learn, XGBoost, Prophet, Random Forest, ARIMA, SARIMA, ensemble methods, GCP, BigQuery, GitHub, time series forecasting

SENIOR DATA SCIENTIST, COPPEL; CDMX – JAN 2022 - JUL 2025

Predictive Model for Expansion of Motorcycle Agencies

Spearheaded a predictive modeling initiative to guide the expansion strategy of Coppel's motorcycle agency network, targeting 542 localities by 2030. Built and validated two machine learning models (one baseline, one enhanced) to predict first-year sales using 117–119 locality-level variables. Integrated external (INEGI, DENUE) and internal (sales history, digital channels) data sources. Delivered a ranked list of high-potential locations and key sales drivers.

Tools: Python (Scikit-learn, XGBoost), SQL, geo data integration, feature engineering, data visualization

National Queue Management Project for Retail Stores

Enhanced the management of waiting queues in retail stores across the nation using clustering algorithms to identify key store characteristics. Incorporated principles from Queueing Theory to create a KPI to gauge queue sizes, optimizing queue management strategies and improving customer flow dynamics on a national scale.

Tools: NumPy, Pandas, Scikit-Learn, Jupyter Notebook, DBeaver, Queueing Theory

Pricing Project for Furniture Category (CENIC-MIT Collaboration)

Cutting-edge collaboration between CENIC and Massachusetts Institute of Technology (MIT) focused on optimizing pricing strategies to enhance profitability in the national furniture market. Deployed final outcome as an intuitive web application. Achieved profitability enhancement ranging from 5% to 20% through strategic implementation of new pricing strategies.

Tools: NumPy, Pandas, Scikit-Learn, Jupyter Notebook, IBM Cloud Pack, IBM CPLEX, SQL, DBeaver, time series analysis and forecasting

Pricing Project for Clothing Category

Built upon MIT suggestions with novel approaches incorporating Feedforward Neural Networks (FNN) and Long Short-Term Memory (LSTM) networks, along with unsupervised algorithms for time series clustering of demand data. Addressed the clothing market nationally.

Tools: Keras, TensorFlow, Azure Data Studio, SQL Server, Matplotlib, Seaborn, time series analysis and forecasting

Business Case: Integration of GANs, Computer Vision, and Social Media Data Analysis in Fashion

Proposed innovative strategy combining Generative Adversarial Networks (GANs), Computer Vision, and social media data analysis to revolutionize clothing design and fashion trend forecasting.

Facial Expression Recognition (FER) Model Development

Led development of FER model for in-store customer analysis using transfer learning techniques with Xception and Inception architectures in TensorFlow and PyTorch frameworks.

Tools: TensorFlow, PyTorch, OpenCV, Google Colab

Computer Vision Automation

Led conceptualization and execution of Computer Vision-driven automation system for internal processes. Developed classifier using transfer learning with MobileNet in TensorFlow, integrated into API used by automation bot.

Tools: TensorFlow, PyTorch, OpenCV, SCRUM

Visual Search Demo on GCP

Led team of four in creating visual search demo leveraging Google Cloud Platform. Implemented visual search functionality to enhance image search and recognition capabilities.

Tools: GCP Cloud Source Repositories, Docker, TensorFlow, PyTorch, OpenCV, Git, Linux, SCRUM, Flask

Statistical Methodology Enhancement for Data Analytics and Supply Chain

Reviewed and enhanced statistical methodology in data analytics and supply chain. Focused on analyzing and rectifying existing methodology for measuring business trials, implementing new statistical tools for more accurate assessment.

Tools: Hypothesis testing, time series clustering, advanced statistical algorithms, hierarchical cluster analysis

Pagerank-Based Personalized Recommendation Algorithm

Crafted algorithm delivering personalized recommendations based on user preferences and behaviors, adapting Pagerank principles. Employed multiprocessing library for handling substantial data volumes with rapid response times.

Tools: PySpark

DATA SCIENTIST, CONSEJO DE LA JUDICATURA; CDMX – NOV 2017 - DEC 2021

Unsupervised Algorithm for Time Series Clustering in Judicial Statistics

Implemented innovative unsupervised clustering algorithm to efficiently allocate human resources based on patterns in time series data. Devised solution that effectively grouped similar time series, facilitating strategic resource allocation. Migrated initial SPSS version to R to leverage advanced data analysis capabilities.

Tools: Microsoft Office, SPSS, R, hierarchical cluster analysis

Text Classification Model for National Census of Federal Justice

Led development of sophisticated text classification model for "National Census of Federal Justice Impartation" to pre-classify federal offenses. Utilized natural language processing techniques and machine learning algorithms to analyze and categorize textual descriptions of reported offenses, achieving accurate and efficient pre-classification.

PROFESSOR, NATIONAL AUTONOMOUS UNIVERSITY OF MEXICO; CDMX – JAN 2012 - JUL 2017

Served as professor at the Faculty of Science, UNAM, designing and conducting diverse courses including Probability, Bayesian Statistics, Mathematics for Biology and Physics, Seminars on Actuarial Applications, Algebra, Stochastic Processes, Mathematical Analysis, Differential Equations, and Risk Theory.

EDUCATION

PhD in Mathematics. 2012-2017

National Autonomous University of Mexico, CDMX

Expected Graduation: AUG, 2026.

M.S. in Science. 2010-2012

National Autonomous University of Mexico, CDMX

Graduated: JUNE, 2012

Bachelor of Mathematics. 2006-2009

National Autonomous University of Mexico, CDMX

Graduated: JAN, 2010

Bachelor of Actuarial Science. 2005-2007

National Autonomous University of Mexico, CDMX

Expected Graduation: AUG, 2026.

CERTIFICATIONS

- Supervised Machine Learning: Regression (Coursera 2024).

- Inferential Statistics (Coursera 2023).
- Object Detection Using Facebook's Detectron2 (Coursera 2023).
- Deep Learning with PyTorch: Image Segmentation (Coursera 2023).
- Image Segmentation with Python and Unsupervised Learning (Coursera 2023).
- Classification with Transfer Learning in Keras (Coursera 2023).

GENERAL SKILLS

Collaborative Team Player | Problem Solving | Effective Communication | Adaptability | Creativity | Solution-Oriented | Teamwork | Analytical Thinking | Innovation | Cross-Functional Collaboration

LANGUAGES

- Spanish (native)
- English (fluent)

SOFTWARE DEVELOPMENT

Apolonio v1. A graphical interface to teach a generalization of the

circle, ellipse, parabola and hyperbola. Laboratory of scientific computation of the

National Autonomous University of Mexico. (Matlab): JAN 2007- DEC 2010. Tools Utilized: *Matlab*.

LECTURES

Reconstrucción booleana de materiales porosos, invited talk in the

DÉCIMA SEMANA INTERNACIONAL DE LA ESTADÍSTICA Y LA PROBABILIDAD,

Puebla, Puebla. JUNE 2017.

Una aplicación de la Teoría Ergódica en la búsqueda de Google in the XLII CONGRESO NACIONAL DE LA SOCIEDAD MATEMÁTICA, Zacatecas, Zacatecas. Oct 2017.

INTERESTS

- Exploring emerging trends in AI, machine learning, and automotive technology.
- Contributing to open-source data science projects.
- Advanced forecasting methodologies and time series analysis